

PolyAgora V8 – Momentum-as-Geometry Architecture

Technical Rewrite Specification for CTO Team

Core Architectural Shift

PolyAgora should no longer treat Momentum as:

- an external benchmark,
- or a competing allocator.

Instead:

Momentum becomes an internal manifold coordinate.

Meaning:

Momentum is absorbed into the geometry layer of the Reference Polygon itself.

This transforms PolyAgora from:

- a topology-aware allocator using external momentum comparisons,

into:

- a topology-aware manifold system where momentum is endogenous to admissibility structure.

Philosophical Principle

Classical momentum assumes:

$$\text{Past Persistence} \rightarrow \text{Future Allocation}$$

PolyAgora V8 instead assumes:

$$\text{Momentum} \subset \text{Admissibility Geometry}$$

Momentum becomes:

- one coordinate of coherence,
- not the steering law itself.

Design Objective

The objective of V8 is:

NOT:

to outperform momentum by rejecting it,

BUT:

to contextualize momentum inside regime geometry.

This allows:

- momentum continuation during stable local-star regimes,
- while suppressing momentum near topology fractures and transition zones.

New Reference Polygon Structure

Suggested initial Polygon dimensions:

Segment	Meaning
M	Momentum Persistence
V	Volatility Stability
L	Liquidity Structure
C	Cross-Asset Coherence
Φ	Mission / Macro Integrity
R	Recoverability / MRTP Capacity

Optional future dimensions:

- convexity,
- crowding,
- dispersion,
- carry,
- geopolitical coherence (VAIDM),
- optionality.

Momentum Segment Definition

Important:
Do NOT use raw trailing returns directly.

Instead define Momentum Persistence segment:

$M_i(t)$

as a normalized persistence field.

Suggested formulation:

$$M_i(t) = z \left(\frac{\text{EMA}_{63}(r_i) - \text{EMA}_{252}(r_i)}{\sigma_i} \right)$$

Alternative variants:

- coherence-adjusted momentum,
- block-relative momentum,
- cross-sectional persistence rank,

- regime-conditioned persistence.

Goal:

Momentum should measure:

- persistence quality,
NOT raw price acceleration.

Architectural Rewrite

OLD SYSTEM

Pipeline:

Returns

- Momentum Scores
- Survival Matrix
- Block Selection
- Allocation

NEW V8 SYSTEM

Pipeline:

Returns / Vol / Liquidity / Macro Signals

- Polygon Segment Construction
- Regime Geometry
- Local-Star Detection
- Soft Admissibility Field
- Runtime Steering
- Allocation

Momentum is now upstream geometry.

Not downstream allocation.

Critical Rewrite Principle

The allocator must NEVER directly allocate from momentum rank alone.

Instead:

Momentum modifies:

- admissibility,
- block coherence,
- local-star stability,
- regime persistence probability.

This is a major architectural shift.

Soft Admissibility Layer

V7.4 showed hard-mask failures.

Therefore V8 must use:

probabilistic manifold weighting.

Required changes:

Replace:

- argmax local-star selection,
- hard admissibility masks.

With:

- softmax block weighting,
- entropy-preserving gating,
- top-k local-star mixtures.

Suggested:

$P(B_k | X_t)$

$$= \frac{e^{\beta S_k}}{\sum_j e^{\beta S_j}}$$

where:

- B_k = block,
- S_k = block coherence score,
- β = geometry rigidity parameter.

Momentum Interaction Logic

Momentum should affect:

1. Block Stability

Strong coherent momentum:

- strengthens stable local stars.

Divergent momentum:

- weakens coherence.

2. Transition Detection

Momentum divergence across segments:

- indicates topology stress,
- possible boundary migration.

This becomes transition signal.

3. Recoverability

Momentum concentration near:

- high volatility,
- low liquidity,
- high crowding,

should REDUCE recoverability score.

This aligns with MRTP.

New Regime Logic

Regimes become:

polygon geometries,

NOT discrete labels.

Meaning:

- regimes emerge continuously,
- rather than being explicitly classified.

This is important.

Runtime Steering Layer

Driver Seat should now operate on:

Control	Meaning
Geometry Rigidity β	Soft vs hard admissibility
Momentum Sensitivity α_M	How strongly persistence shapes geometry

Recoverability λ_R	M RTP weighting
Convexity Preference κ	Optionality vs carry
Transition Sensitivity τ	Early topology fracture detection

External Polygon Integration

VAIDM / ADD / Buffett Polygon should operate as:

admissibility deformers.

NOT direct allocators.

Meaning:
they modify:

- geometry curvature,
- zone transitions,
- block admissibility probabilities.

Especially near:

- boundary zones,
- topology fractures,
- macro stress.

Critical Insight from V7.4

V7.4 failure demonstrated:

geometric rigidity destroys diversification.

Therefore:

The manifold must remain:

- deformable,

- probabilistic,
- partially ambiguous.

This is not a bug.

This is likely a fundamental property of admissible systems.

Oracle Benchmark Interpretation

Momentum approximates:

- local persistence.

Oracle approximates:

- future admissibility.

PolyAgora V8 attempts:

- partial inference of admissibility transitions BEFORE price collapse.

This should remain the scientific framing.

Avoid framing system as:

- "predictive oracle."

Instead:

- "topology-aware maneuverability engine."

Immediate CTO Deliverables

Priority 1

Rewrite Polygon construction layer:

- momentum becomes segment coordinate.

Priority 2

Implement soft admissibility weighting.

Priority 3

Remove hard block masks entirely.

Priority 4

Build regime geometry visualizer.

Priority 5

Test:

- V6.3,
- V7.3,
- V8 soft geometry, head-to-head.

Suggested Initial Experiments

Experiment A

Momentum:

- as allocator.

vs

Momentum:

- as polygon segment.

Experiment B

Hard local-star masks

vs

soft top-k mixtures.

Experiment C

Transition-period analysis:

- 2020,
- 2022,
- volatility spikes,
- crisis transitions.

Measure:

- recoverability,
 - DD,
 - turnover,
 - convexity preservation.
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Final Strategic Direction

PolyAgora should evolve toward:

a semantic risk geometry operating system.

Not:

- a momentum engine,
- nor a prediction engine.

But:

- a runtime admissibility architecture,
where momentum becomes:
- one manifestation of local manifold coherence.